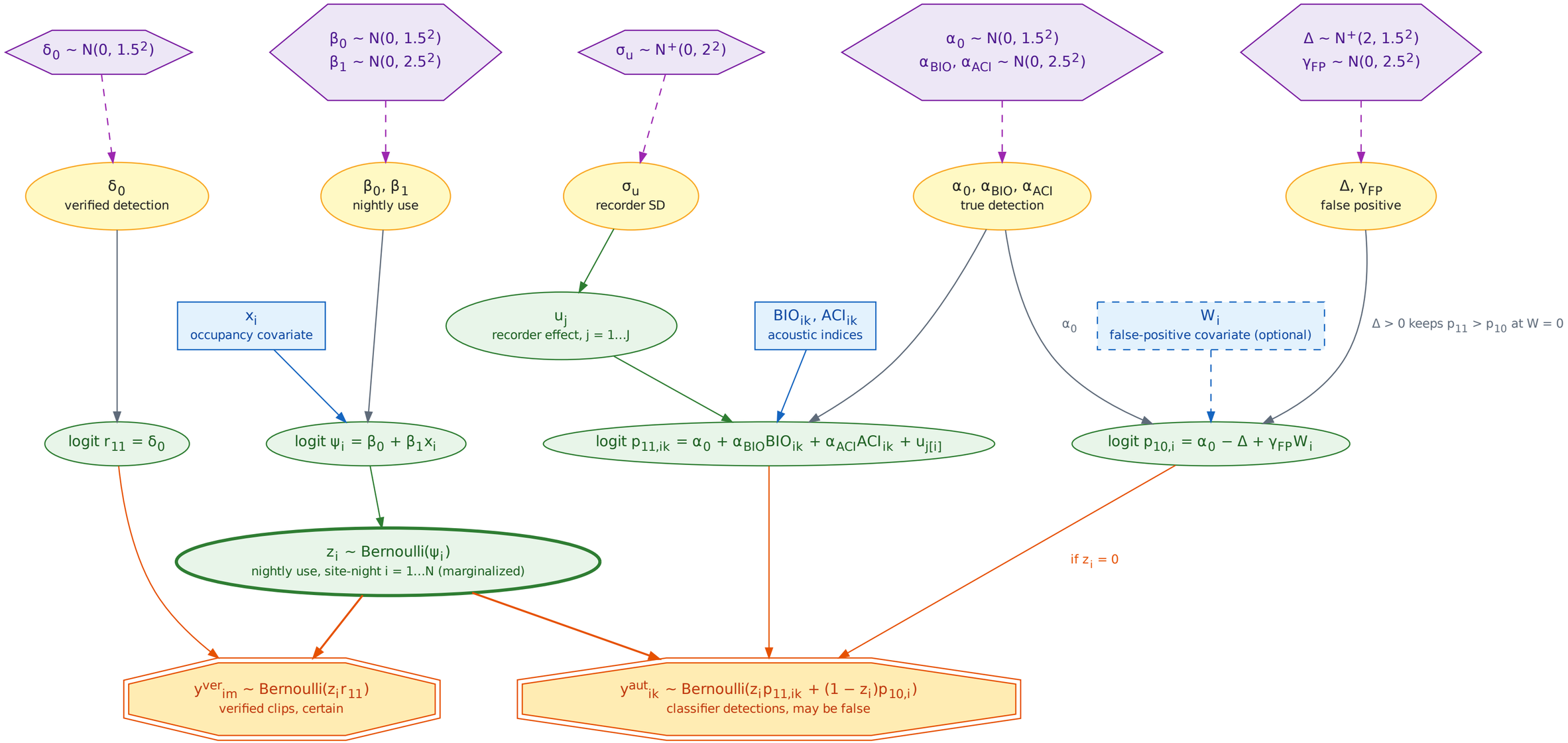
$\delta_0 \sim N(0, 1.5^2)$
$\beta_0 \sim N(0, 1.5^2)$
$\beta_1 \sim N(0, 2.5^2)$
$\sigma_u \sim N^+(0, 2^2)$
$\alpha_0 \sim N(0, 1.5^2)$
$\alpha_{BIO}, \alpha_{ACI} \sim N(0, 2.5^2)$
$\Delta \sim N^+(2, 1.5^2)$
$\gamma_{FP} \sim N(0, 2.5^2)$
δ_0
verified detection
β_0, β_1
nightly use
σ_u
recorder SD
$\alpha_0, \alpha_{BIO}, \alpha_{ACI}$
true detection
Δ, γ_{FP}
false positive
x_i
occupancy covariate
u_j
recorder effect, j = 1…J
BIO_{ik}, ACI_{ik}
acoustic indices
α_0
W_i
false-positive covariate (optional)
$\Delta > 0$ keeps $p_{11} > p_{10}$ at $W = 0$
logit $r_{11} = \delta_0$
logit $\psi_i = \beta_0 + \beta_1 x_i$
logit $p_{11,ik} = \alpha_0 + \alpha_{BIO} BIO_{ik} + \alpha_{ACI} ACI_{ik} + u_{j[i]}$
logit $p_{10,i} = \alpha_0 - \Delta + \gamma_{FP} W_i$
$z_i \sim$ Bernoulli(ψ_i)
nightly use, site-night i = 1…N (marginalized)
if $z_i = 0$
$y^{ver}_{im} \sim$ Bernoulli($z_i r_{11}$)
verified clips, certain
$y^{aut}_{ik} \sim$ Bernoulli($z_i p_{11,ik} + (1 - z_i) p_{10,i}$)
classifier detections, may be false